# Customer Shopping Behavior Analysis

Analyzing transactional data to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

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# Project Overview & Dataset Summary

## Project Goal

Uncover insights from 3,900 purchases across various product categories to inform strategic business decisions.

## Key Features

- Customer demographics (Age, Gender, Location, Subscription Status)
- Purchase details (Item, Category, Amount, Season, Size, Color)
- Shopping behavior (Discount, Promo, Previous Purchases, Frequency, Review, Shipping)

## Dataset Snapshot

- Rows: 3,900
- Columns: 18
- Missing Data: 37 values in Review Rating

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# Exploratory Data Analysis in Python

We initiated the analysis with data preparation and cleaning using Python.

1

### Data Loading & Exploration

Imported dataset with pandas and used `df.info()` and `.describe()` for initial checks.

2

### Missing Data Handling

Imputed missing 'Review Rating' values using the median rating per product category.

3

### Feature Engineering

Created 'age_group' and 'purchase_frequency_days' for deeper analysis.

4

### Database Integration

Loaded the cleaned DataFrame into PostgreSQL for subsequent SQL analysis.

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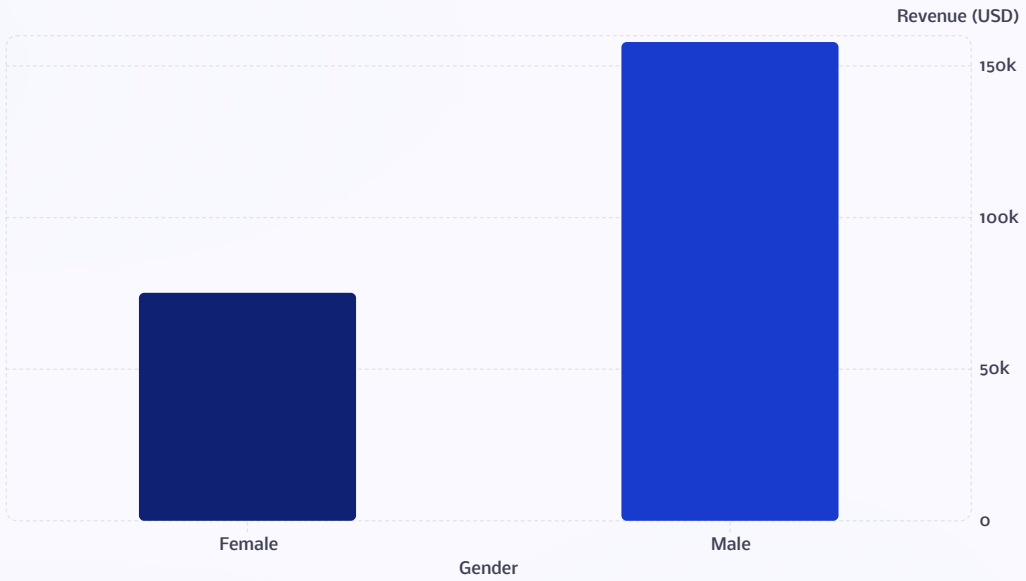
# Data Analysis using SQL: Key Insights

Structured analysis in PostgreSQL answered critical business questions.

## Revenue by Gender




## High-Spending Discount Users

Identified 839 customers who used discounts but spent above average.

## Top 5 Products by Rating

| | |
|---|---|
| Gloves | 3.86 |
| Sandals | 3.84 |
| Boots | 3.82 |
| Hat | 3.80 |
| Skirt | 3.78 |

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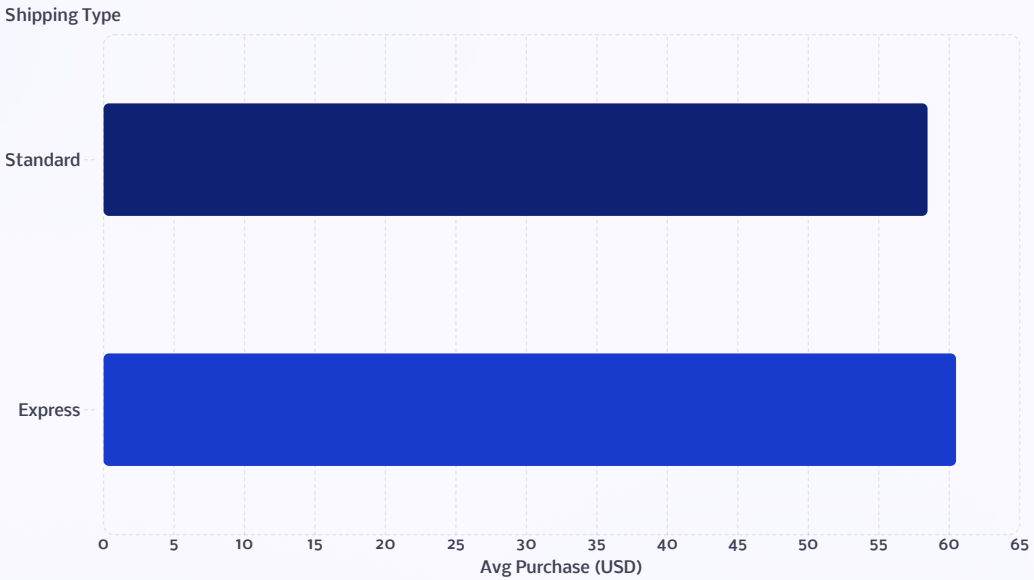
# SQL Analysis: Shipping & Subscriptions

## Shipping Type Comparison




## Subscribers vs. Non-Subscribers

| Subscription Status | Total Customers | Avg Spend | Total Revenue |
|---|---|---|---|
| Yes | 1053 | 59.49 | 62645.00 |
| No | 2847 | 59.87 | 170436.00 |

Non-subscribers contribute more total revenue, but average spend is similar.

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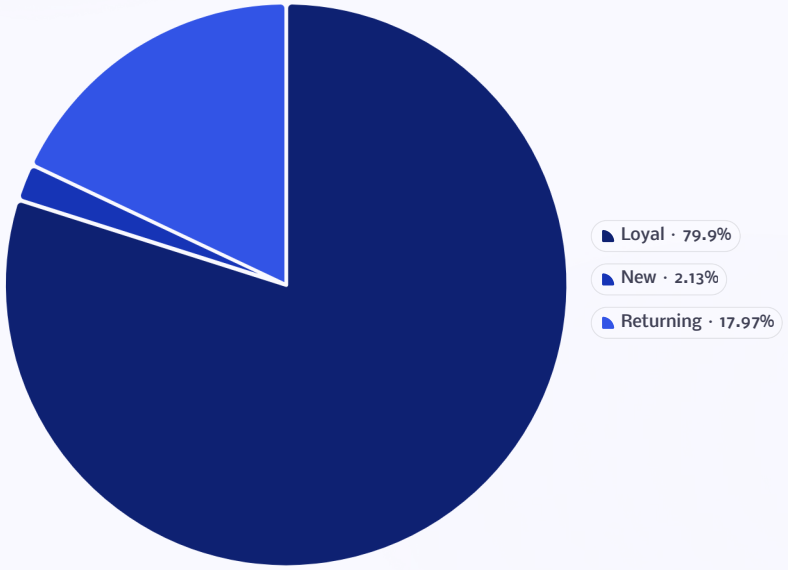
# SQL Analysis: Product & Customer Segmentation

## Discount-Dependent Products

| Item Purchased | Discount Rate (%) |
|---|---|
| Hat | 50.00 |
| Sneakers | 49.66 |
| Coat | 49.07 |
| Sweater | 48.17 |
| Pants | 47.37 |

## Customer Segmentation

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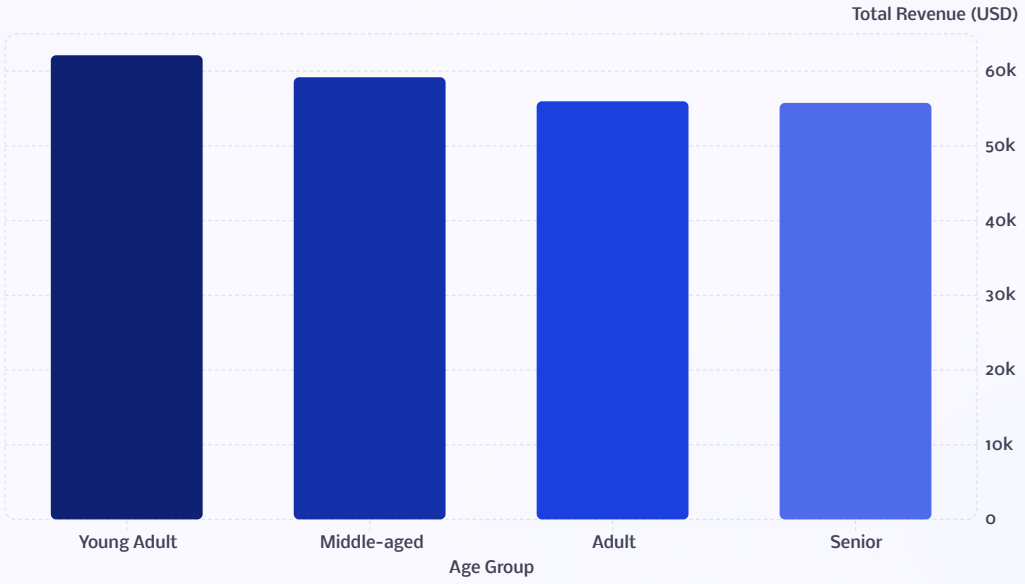
# SQL Analysis: Top Products & Age Groups

## Top 3 Products per Category

| Category | Item Purchased | Total Orders |
|---|---|---|
| Accessories | Jewelry | 171 |
| Accessories | Sunglasses | 161 |
| Clothing | Blouse | 171 |
| Clothing | Pants | 171 |
| Footwear | Sandals | 160 |
| Outerwear | Jacket | 163 |

## Revenue by Age Group

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# Dashboard in Power BI: Visualizing Insights

An interactive Power BI dashboard provides a visual summary of customer behavior, including subscription status, revenue by category, and sales by age group.

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# Key Business Recommendations

### Boost Subscriptions

Promote exclusive benefits to increase subscriber base.

### Customer Loyalty Programs

Reward repeat buyers to foster loyalty and retention.

### Review Discount Policy

Optimize discounts to balance sales boosts with margin control.

### Product Positioning

Highlight top-rated and best-selling products in marketing campaigns.

### Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.

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